

Sreekumar Kallingapuram Suresh

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Portfolio: <https://sreeku1993.github.io/sreekumar.github.io/>

Professional Summary

AI Engineer with an MSc in Artificial Intelligence and 4+ years of experience across software engineering, QA automation, and AI application development. Experienced in building Agentic AI systems, LLM-powered applications, RAG pipelines, AI assistants, and workflow-driven automation using Python, FastAPI, LangGraph, MCP, LangChain, LangSmith, Google Gemini, and modern AI frameworks. Skilled in designing scalable backend services, vector search architectures, AI evaluation pipelines (RAGAS, LLM-as-a-Judge), and deploying production-ready AI solutions. Proven experience designing and building end-to-end AI products including healthcare agents, legal research assistants, and career intelligence platforms.

Skills

Backend & Application Development

Python, FastAPI, Flask, REST APIs, Next.js, TypeScript, Java, JavaScript

Agentic AI, LLMs & GenAI

Agentic AI, LangGraph, Model Context Protocol (MCP), LangChain, Google Gemini, LLMs, AI Agents, Prompt Engineering, Tool Calling, Multi-Agent Workflows, Conversational AI

RAG & AI Evaluation

Retrieval-Augmented Generation (RAG), Semantic Search, Embedding Models, Vector Search, Hybrid Retrieval, BM25, Reciprocal Rank Fusion (RRF), Reranking, RAGAS, LLM-as-a-Judge, LangSmith, Automated Evaluation Pipelines, Benchmark Dataset Creation, SentenceTransformers, OCR, PDF processing

Machine Learning & Computer Vision

PyTorch, TensorFlow, Scikit-learn, OpenCV, YOLO

Databases & Data Processing

MySQL, SQL Server, NoSQL (basic), Supabase, Pandas, Data preprocessing

Cloud & DevOps

Docker, Kubernetes, GCP (Cloud Run), Vercel, Git, Jenkins, CI/CD, Render

Automation & Testing

AI System Testing, Prompt Testing, Automated Evaluation Pipelines, Selenium WebDriver, Cypress, Playwright, TestNG, Postman, UFT

Tools & Platforms

VS Code, Jira, Confluence, Xray, Google App Scripts, Android Studio

Security & AI Safety

Hallucination Reduction Strategies (RAG validation), Privacy-aware Prompting, LLM Output Safety Filtering(Guardrails), Grounded Response Validation

Development Methodologies

Agile / Scrum, Software Design, Test Strategy, QA Automation

Professional Experience

Freelance AI Engineer (AI Consulting & Product Development)

Independent – Thrissur, India | Jan 2026 – Present

- Designing and developing production-grade AI applications across healthcare, legal, and career intelligence domains.
- Building Agentic AI systems using LangGraph, MCP, LangSmith, FastAPI, and modern LLM frameworks.
- Developing Retrieval-Augmented Generation (RAG) applications with hybrid retrieval, vector databases, evaluation pipelines, and observability tooling.
- Delivering end-to-end AI solutions including architecture design, backend development, workflow orchestration, deployment, monitoring, and evaluation.

Key Projects:

1. AI-Powered Clinic Assistant (Agentic AI | LangGraph + MCP)
2. AI Legal Assistant (Advanced RAG)
3. Career Intelligence Platform (GenAI SaaS)

Technical Team Leader

Planthood – London, UK | May 2024 – August 2025

- Designed and integrated AI-assisted operational workflows using LLMs and historical production data to generate automated daily status reports, management summaries, and operational insights.
- Developed an Android-based operational platform integrated with Google Sheets to capture real-time task execution, issue reporting, staff activity, and workflow timelines.

- Built backend automation and data-processing pipelines using Python, Django, Google Apps Script, and cloud services for dispatch records, stock tracking, quality checks, and KPI metrics.
- Implemented predictive estimation and operational analytics workflows using historical task data to forecast completion timelines, detect workflow risks, and generate live performance dashboards deployed on Google Cloud Run using Docker.
- Coordinated cross-functional teams to optimize operational efficiency and reduce manual processes.

Programmer Analyst

Cognizant – Kochi, India | May 2021 – Dec 2021

- Contributed to testing and validation of an AI-based voice biometrics system, ensuring accuracy, functionality, and reliable user workflows.
- Performed API validation and backend data verification using Postman and SQL queries on cloud-hosted systems.
- Developed and maintained automation workflows to improve test efficiency and reduce manual validation efforts.
- Collaborated with developers and analysts to troubleshoot issues, validate system behavior, and support stable feature releases.

Test Engineer

Infosys Ltd. – Mysore, India | Aug 2018 – May 2021

- Developed and maintained automation frameworks using Selenium, UFT.
- Implemented CI/CD pipelines to automate testing and deployment workflows.
- Conducted end-to-end testing of e-commerce platforms including checkout flows and payment gateways.

Education

MSc in Artificial Intelligence – University of East London | Jan 2022 – May 2023

Master of Computer Applications – Anna University | Aug 2015 – Jun 2018

Bachelor of Computer Applications – Christ College | Aug 2011 – Aug 2014

Key AI Projects

AI-Powered Clinic Assistant (Agentic AI | LangGraph + MCP)

- Engineering a production-grade agentic healthcare system using LangGraph and Model Context Protocol (MCP), with specialized agents for appointment scheduling, patient management, document retrieval, and clinic operations.
- Developing graph-based workflow orchestration with persistent state management, multi-turn conversational reasoning, tool-calling workflows, and human-in-the-loop approval checkpoints.

- Building MCP tool servers exposing healthcare operations including patient lookup, appointment booking, availability search, rescheduling, cancellation, and secure document access.
 - Designing scheduling intelligence covering doctor availability, clinic schedules, holidays, breaks, leave management, and appointment conflict detection.
 - Integrating LangSmith observability for workflow tracing, debugging, evaluation, and agent performance monitoring across complex execution paths.
 - Developing FastAPI backend services, PostgreSQL data models, and workflow-driven business logic to support scalable healthcare operations.
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AI Legal Assistant (Advanced RAG)

- Developed an AI-powered legal research assistant using Groq (Llama 3.3 70B), Qdrant Vector Database, BM25 retrieval, and Reciprocal Rank Fusion (RRF).
 - Ingested and indexed over 32,000 legal chunks from the Constitution of India and statutory law datasets.
 - Implemented hybrid retrieval combining semantic vector search and keyword search with query-aware routing for Articles, Sections, IPC, and BNS references.
 - Designed retrieval optimization pipelines using metadata filtering, hybrid ranking, and contextual reranking techniques.
 - Integrated LangSmith tracing for end-to-end observability, retrieval debugging, and performance analysis.
 - Evaluated system quality using RAGAS metrics including Faithfulness, Answer Relevancy, Context Precision, and Context Recall.
 - Built and deployed a full-stack web application using Next.js, TypeScript, and Vercel.
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Career Intelligence Platform (GenAI SaaS)

- Designed and developed an AI-powered career intelligence platform that analyzes resumes, assessments, and user profiles to generate personalized career transition recommendations.
 - Built a multi-stage LLM workflow using Google Gemini for resume analysis, role matching, skill-gap identification, career path recommendations, and structured learning roadmap generation.
 - Developed personalized AI guidance workflows combining user profiles, assessment results, learning activity, and career context to deliver tailored recommendations and conversational support.
 - Implemented resume parsing, structured information extraction, skill categorization, and recommendation pipelines to support career planning and decision-making.
 - Built the full-stack application using Next.js, Supabase, FastAPI, and Vercel, including authentication, user management, storage, and scalable backend services.
 - Designed the platform architecture for future integration of vector-based retrieval, contextual memory, payment processing, and multi-model AI capabilities.
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